

SAMEER ARJUN SATHEESH

+1 240-424-8528 | sameerarjuns@outlook.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [YouTube](#) | Salt Lake City, UT

Robotics and Automation Engineer experienced deploying AGV/AMR fleets, industrial controls, and mechanical systems from concept through production. Combines firmware, path-planning, and systems-integration expertise with a track record of measurable reliability & cost improvements.

TECHNICAL SKILLS

Robotics & Autonomy: ROS, ROS2, SLAM, Path Planning (RRT*, MAPP), Sensor Fusion, OpenCV, Kollmorgen NDC

Software & Control: Python, C++, Embedded C++, MATLAB, PLC (Ladder Logic), Git, JIRA, Docker, Linux, TwinCAT3

Mechanical & Simulation: Siemens NX, CATIA, ANSYS (FEA), SolidWorks, Gazebo, AnyLogic, GD&T, DfM

Protocols & Diagnostics: CAN Bus, XML/JSON, TCP/IP, Wireshark, API Integration, Elasticsearch

EDUCATION

University of Maryland, College Park

Aug 2022 - May 2024

Master of Engineering in Robotics; CGPA: 3.78/4.00

Visvesvaraya Technological University, India

Aug 2016 - Aug 2020

Bachelor of Engineering in Mechanical Engineering; CGPA: 8/10

WORK EXPERIENCE

Toyota Material Handling / The RAYMOND Corp.

Jul 2024 - Present

Applications Engineer II

- Architected and deployed 26 autonomous forklift systems using Kollmorgen NDC Solutions, cutting manual guidance downtime and enabling reliable 24/7 AGV operation across active warehouse sites
- Designed localization and routing models for site-specific layouts, scaling navigation to mixed AGV/AMR fleets without collision incidents or re-routing failures
- Diagnosed and resolved system failures using Wireshark and CAN bus analysis, reducing fault recurrence and improving fleet up time
- Integrated AGV platforms with WMS and ERP systems via API, automating task dispatch and enabling real-time fleet tracking for customer operations teams
- Standardized technical documentation and led onboarding training, cutting ramp-up time for new engineers and customer technical staff

Stanley Black and Decker Inc.

Jun 2023 - Aug 2023

Electro-Mechanical Engineering Intern

- Engineered SBS trigger modules for universal power tool platforms using CATIA, enabling consistent mechanical fit and assembly across platforms
- Root-caused DeWalt impact drill failures and co-introduced polymer-based capacitors to SBS architectures, improving module reliability and electrical performance
- Supported the CraftsMan EV Charger pipeline for an internal innovation challenge, informing product strategy and market-expansion planning

MOLEX India Business Services Pvt. Ltd.

Mar 2021 - Jul 2022

Graduate Engineering Trainee

- Designed and analyzed power and signal connector systems in Siemens NX, enabling reliable performance in high-performance packaging applications
- Implemented FEA-based mechanical drop test simulations for connector design validation, saving \$1M annually and accelerating time-to-market
- Validated liquid cooling technologies for data center high-power and signal connectors, increasing current-carrying capacity by 120%
- Presented technical results at the 2022 Open Compute Project Global Conference

PROJECTS

Autonomous 4-Wheeled Mobile Manipulator | Python, Raspberry Pi, Arduino, OpenCV - [YouTube](#)

- Built a differential drive robot with a parallel-jaw gripper and real-time CV pipeline for target recognition and depth estimation; fused IMU, ultrasonic, and encoder data for closed-loop localization and control
- Implemented robust sensor fusion using IMU (Inertial Measurement Unit), ultrasonic sensors, and wheel encoders to achieve localization and closed-loop control

Swarm Robotics for Industrial Applications | C++, ROS2 Humble, Gazebo - [GitHub](#)

- Architected a swarm system implementing Multi-Agent Path Planning (MAPP) for 20 TurtleBots, achieving coordinated navigation; developed the communication stack and obstacle-avoidance logic using Agile methodology

ACHIEVEMENTS

- Best Research Paper Award, Make in India Research Paper Contest, Government of India;
- Winner, Super Float Idea Challenge, MOLEX India
- Certificate of Appreciation, MOLEX-KOCH Emergency Response Team (COVID-19 volunteer work)